Objective Assessment of Surgical Skill Using Artificial Intelligence Hand Tracking in Cardiothoracic Training: A Feasibility Study.

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Abstract

Measuring surgical competency is essential for surgical residents to ensure patient safety. Traditional assessment tools rely on subjective evaluation. This study evaluated whether artificial intelligence (AI)-based hand tracking can more objectively distinguish between levels of surgical competency and predict surgical years of experience versus traditional assessments. A total of 44 participants, including medical students, surgical residents, and surgical consultants, performed transcutaneous suturing, intracutaneous suturing, and surgical knot tying. Videos of intracutaneous suturing were scored using the objective structured assessment of technical skills (OSATS). Hand movements were analysed using AI tracking software to extract coordinates to measure velocity, pathlength, and jerk. Linear regression models predicted experience years using procedural time and OSATS in combination with hand tracking metrics. Hand tracking metrics varied mainly between medical students and more experienced groups. Traditional assessment tools (procedural time, OSATS) could predict experience years during training, with an adjusted coefficient of determination (R2) ranging from 0.537 to 0.638, dependent on procedure type. Hand tracking variables identified multiple significant predictors for years of experience, with an adjusted R2 of 0.540-0.712, which outperformed the traditional tools in each procedure. Combining all assessment tools (time, OSATS, and hand tracking) gave the best predictive value, with an adjusted R2 ranging from 0.540 to 0.809, with velocity, pathlength, jerk, and acceleration as significant predictors. AI-based hand tracking provides a new method for objective, reproducible measures of surgical skills. Incorporating hand tracking metrics enhances prediction of surgical experience, and supports standardized as well as objective evaluation of skills assessment in surgical training.

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